Sahaj Bhadja

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Summary

Full Stack Software Engineer with 2+ years delivering **React/TypeScript products**, scalable backend services, financial workflows, and production AI systems using Node.js, Python, PostgreSQL, MongoDB, testing, CI/CD, and secure cloud infrastructure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Software Engineering, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Data Structures, Database Management, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected a secure multi-tenant platform using Azure SQL and Blob Storage with normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query optimization.
- Restructured fault-tolerant asynchronous AI orchestration using FastAPI background workers, processing 1K+ jobs with explicit state transitions and failure isolation while tripling throughput under concurrent production workloads with reliable recovery.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Built responsive React interfaces and Python services for uploads, grading history, configuration, and live feedback, reducing turnaround 60% while supporting reliable workflows used by 150+ instructors across multiple courses.
- Engineered multimodal AI workflows using GPT-4o, Claude, Qwen-VL, and OCR, evaluating 10K+ daily submissions with 90%+ consistency while validating outputs, tracking usage, and handling model failures under production load.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Optimized scalable API integrations through object-oriented design, filtering, pagination, indexing, and caching, improving response times 30%, increasing data-pipeline efficiency 68%, and reducing technical debt 20% across production workloads.
- Engineered cross-platform software with automated builds and 20+ unit and integration test suites, validating releases across three operating systems while collaborating with Microsoft and GitHub engineering stakeholders.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Built responsive React workflows and eight modular MERN services for authentication, permissions, billing, resources, customers, leads, and notifications, synchronizing 2M+ records through REST APIs and WebSockets across production workflows.
- Automated Docker deployments through Jenkins CI/CD with unit, integration, and end-to-end validation, achieving 95%+ successful builds while integrating monitoring and secure checks across staging and production environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built responsive React and React Native booking experiences across web, Android, and iOS, integrating Flask APIs, Firebase synchronization, and payments while maintaining sub-200-millisecond responses across 50K+ records.
- Designed end-to-end user workflows, mapped 12+ integration points, and presented architecture to client stakeholders, reducing design errors 25% through iterative feedback and cross-functional implementation alignment under evolving requirements.

Skills

Full Stack: TypeScript, JavaScript, Node.js, React.js, Next.js, HTML, CSS, TailwindCSS, Python, Java, FastAPI, Flask, Express.js, REST APIs, WebSockets
Data & Cloud: PostgreSQL, MySQL, MongoDB, NoSQL, Firebase, Azure SQL, AWS, Azure, GCP, Docker, Kubernetes, Linux, JSON, Data Modeling
AI Development: LLM Systems, GPT-4o, Claude, Qwen-VL, OCR, Prompt Chains, Model Evaluation, Semantic Search, Vector Search, Transformers, PyTorch, Workflow Automation
Engineering: Git, GitHub, Jenkins, GitHub Actions, CI/CD, JIRA, Postman, Unit Testing, Integration Testing, End-to-End Testing, Code Reviews, Debugging, Monitoring, Performance Optimization, System Design, Agile/Scrum, Secure Coding, Cross-Functional Collaboration

Projects

FinNexus Live | *Next.js, TypeScript, MongoDB, Plaid, WebSockets, Kubernetes*

- Built a responsive fintech platform supporting 50K+ users and 1M+ transactions with secure bank linking, real-time synchronization, advanced filtering, tokenized access, and scalable Kubernetes deployment automation for reliable operations.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built an AI-assisted document platform with secure ingestion, semantic retrieval, summarization, and collaborative workflows, maintaining 95% query accuracy across 50K+ documents while reducing review time 60% through dependable automation.

PiXMod Live | *Next.js, Angular.js, Stripe, MongoDB, CLIP*

- Built a responsive SaaS interface with Stripe subscriptions, scalable MongoDB storage, semantic image search, segmentation, and generative editing workflows, improving premium conversion 18% across an AI-powered product experience.